EMERGING TECHNOLOGIES | WEEK 6

Finding Unusual Expense Reports with Linear Regression

Meridian Fixtures Inc. | T

ravel Expense Reports, January–June 2026 | Estimated time: 1 hour 20 minutes

The Situation

Meridian Fixtures' employees submitted 129 travel expense reports between January and June 2026. The accounts payable team reviews each one by hand, and the Controller wants to know whether a simple model could point reviewers to the reports most worth a second look.

Your team's job: build a linear regression that predicts a trip's total cost from its length in days, use it to flag reports that don't fit, investigate each flag, and then test what the model misses. You'll finish by fact-checking an AI-drafted audit memo before it reaches the Controller, and then generating and checking one of your own.

What You'll Need

- `travel_expenses_STUDENT.csv`: the expense report data.
- `Week6_LinearRegression_Outliers_STUDENT.ipynb`: the notebook. Open it in Jupyter.
- An AI assistant (Claude, ChatGPT, Copilot, or similar) for Task 6, Part B.
- The answer sheet at the end of this document.

> **You don't need to write code from scratch.** The notebook has the code ready. In four places you'll replace the word `FILL_IN` with the right value. Hints are in the comments. Run each cell with **Shift + Enter**, in order from top to bottom.

Meridian's Travel Policy (Relevant Excerpts)

1. No single trip may exceed **$8,000** without CFO pre-approval. No pre-approvals were granted from January to June 2026.
2. Trip reports must have a positive total. Refunds and reversals are processed separately and should never appear as a trip report.
3. The number of days claimed must match the trip's start and end dates (both days count).
4. International travel is permitted with manager approval. International airfare typically costs more than domestic airfare.

The Data

Column	What it means	Example
`report_id`	Unique expense report number	TR-2001
`employee`	Who submitted the report	S. Patel
`destination`	Where the trip was	Phoenix, AZ
`purpose`	Reason for the trip	Customer visit

Column	What it means	Example
`trip_start` / `trip_end`	First and last day of the trip	2026-02-28
`days_claimed`	Number of days the employee entered on the report	3
`total_cost`	Total dollars claimed for the trip	1158.78

Time Plan

Task	Time	What you'll do
0. Setup	5 min	Open the notebook, upload the CSV, run the first cells
1. Look before you model	5 min	Summary statistics and a scatter plot
2. Fit a line to all the data	10 min	Run your first regression and interpret it
3. Remove obvious errors and refit	10 min	Apply the policy rules, fit again, compare
4. Use residuals to find outliers	20 min	Flag unusual reports and investigate each one
5. Adversarial test	10 min	Find what the model missed
6. Fact-check an AI-drafted memo	15 min	Find the errors in an AI memo, then test your own AI
Wrap-up discussion	5 min	Share findings with the class

Tasks

The notebook walks you through each task step by step, and the questions appear in the notebook too. Type your answers in the notebook's ✍️ cells or on the answer sheet below, whichever your instructor prefers.

Task 0: Setup (5 min)

Open the notebook. In Colab, click the folder icon on the left and upload `travel_expenses_STUDENT.csv`. Run the first two code cells. You should see "Rows: 129" and the first ten reports.

Task 1: Look Before You Model (5 min)

Run the summary statistics and scatter plot cells, then answer:

1. Look at the minimum and maximum `total_cost`. Does anything look impossible or suspicious for a business trip? Why?

   The minimum and maximum both look suspicious. The minimum is -$1,186.40, which is impossible for a business trip because it shouldn't have a negative cost. The maximum is $13,482.50 compared to the 75th percentile of $2,027.02 is more than 6 times the max and far from the other reports.

2. Describe the overall pattern in the scatter plot in one sentence.

   The overall pattern in the scatter plot forms a roughly linear, upward-sloping band, from about $700 for 1-day trips to about $2,500 for 8-day trips, with a few reports that sit well off that band and trip cost generally rises with trip length.

3. Which points stand out just by eye?

The most extreme point is a trip of about 3 days costing about $13,500, which is far above everything else on the chart. The second is a trip of about 2 days with a cost of about -$1,200, which falls below zero. A 9-day trip costing only about $750 also stands out, because it costs about the same as a 1-day trip and is suspiciously cheap. The 18-day trip at about $5,700 sits far to the right of all the other trips. It may not be an expense problem, but it is unusual in length and will pull strongly on the fitted line. A 1-day trip at about $2,500 and a 2-day trip at about $2,950 are also noticeably more expensive than other trips of the same length. Finally, a 5-day trip at about $3,700 and a 6-day trip at about $4,200 sit above the main band, though less dramatically than the others.

Task 2: Fit a Line to All the Data (10 min)

Fill in the two column names (x is the driver, y is what you're predicting), run the regression, and plot the line.

1. Explain the slope in plain business language ("each extra day of travel adds about $___").

   Each extra day of travel adds about $256 to the total cost of a trip, on average. For example, the model predicts a 5-day trip to cost about $256 more than a 4-day trip. This is an average across all reports, so individual trips can be well above or below it.

2. Explain the intercept. What real costs might it represent, even for a very short trip?

   The intercept of $654.15 is the model's predicted cost for a trip of zero days, so it represents the fixed cost of taking any trip at all. Even a very short trip includes costs that don't depend on how long you stay, such as airfare or train tickets, rideshares or airport parking, baggage fees, and conference or event registration. A zero-day trip doesn't exist in the data, so the intercept is best read as a baseline for the line, not a literal prediction.

3. Is R-squared high or low? Looking at the chart, what might be dragging it down?

   An R-squared of 0.199 is low. Trip length explains only about 20% of the variation in trip cost, which leaves about 80% unexplained. That is weaker than the scatter plot suggests, because most reports form a tight upward band. A few unusual reports are dragging it down. The $13,500 report for a 3-day trip is the biggest one, since it sits so far from the line that it adds an enormous amount of unexplained variation on its own. The negative cost report, the 9-day trip costing only about $750, and the short trips with unusually high costs add to this. The 18-day trip also sits far out on the x-axis, where it can tilt the line.

Task 3: Remove the Obvious Errors and Refit (10 min)

Fill in the policy cap from the travel policy, remove reports that break the rules, and fit the regression again.

1. Complete the comparison table on the answer sheet.

   On Answer Sheet

2. How many reports were removed? What happened to R-squared, and why could so few reports make that much difference?

   The policy check removed 2 reports, taking the data from 129 rows to 127. R-squared rose from 0.199 to 0.677, so trip length now explains about 68% of the variation in cost instead of about 20%. Two reports made that much difference because linear regression fits the line by minimizing squared errors, which means a point's influence grows with the square of its distance from the line. The $13,482.50 report sits about $12,000 above where the model predicts a 3-day trip should land (about $1,400). Squaring that distance gives a number so large that this one report accounts for most of the unexplained variation in the whole data set. The -$1,186.40 report adds a smaller but still sizable error of about $2,400 below the line. Once both are removed, the remaining points hug the line closely, so most of the variation is

explained. The slope and intercept moved only modestly (the slope up by about $12 and the intercept down by about $124), because the $13,500 report pulled the intercept up and the other points mostly kept the slope in place.

3. For each removed report, what's the most likely explanation?

   TR-2120 (S. Patel, Atlanta, 2 days, -$1,186.40): A typical 2-day trip costs roughly $1,000, so this amount is about the right size but the wrong sign. The most likely explanation is a sign error or a refund or credit entered as an expense, for example a canceled flight or a reimbursement recorded in the wrong direction. The policy also notes that refunds are handled in a different system, which supports this reading.

   TR-2128 (R. Osei, Dallas, 3 days, $13,482.50): A typical 3-day trip costs about $1,200 to $1,500, so this is roughly 10 times larger. The most likely explanation is a data entry error, such as a misplaced decimal point (for example, $1,348.25 entered as $13,482.50, which would be a normal figure for 3 days). A duplicate charge or a non-trip expense coded to the trip is also possible. Since no CFO pre-approval was granted, it can't be a legitimate expense either way.

Task 4: Use Residuals to Find the Outliers (20 min)

A **residual** is actual cost minus predicted cost. Fill in the standard-deviation multiplier (2), and the notebook will flag every report more than 2 standard deviations from the line.

1. Complete the investigation table on the answer sheet for every flagged report. Use all the columns (destination, purpose, trip dates), not just the residual.

   On Answer Sheet

2. Is every flagged report a problem? Find at least one that might be legitimate, and explain what the model doesn't know.

   No, every flagged report is not a problem and TR-2035 might be a legitimate report. The model uses only the number of days to predict cost, so it doesn't know that Monterrey, Mexico is an international trip with a costlier flight. TR-2081 could also be legitimate, because the model doesn't know that a Sales training trip might include a registration fee. In general, the model knows nothing about destination, purpose, or what the money was spent on, so a large residual means unusual for its length, not wrong.

3. Change the multiplier to 3 and re-run. Does the list change? What's the trade-off between a lower and higher cutoff for an audit team?

   Changing the multiplier to 3 widens the band to $1,266.33 above and below the line, but the flagged list does not change. Every flagged report has a residual well beyond that cutoff. The smallest is TR-2084 at $1,701.86, so even the least extreme report is still about $435 outside the wider band. These five reports are clear outliers, not borderline cases that depend on where the cutoff is set. The trade-off for an audit team is workload against the risk. A lower cutoff (2 standard deviations) flags more reports, so it is more likely to catch every real problem, but it also produces more false alarms that cost reviewers time. A higher cutoff (3 standard deviations) gives a shorter list that is more likely to contain real problems, but it can miss moderately unusual reports, such as a trip that is $1,000 over its expected cost. Which setting is better depends on how costly a missed error is compared with the cost of reviewing extra reports. Here, the list suggests either cutoff would send the same five reports to the auditors.

4. Find the longest trip on the chart. Is it flagged? Why isn't an unusual number of days automatically an outlier?

   The longest trip in the data is the 18-day trip at about $5,700 on the far right of the chart. It is not flagged: it is shown as a blue "Within range" dot, sitting inside the dashed band, close to the regression line (predicted about $5,360, so a residual of only about $380). An unusual x-

value does not make a report an outlier, because an outlier is defined by its distance from the line, which is the residual, in the y direction. This trip is far from the other trips in length, but its cost is about what the pattern predicts for 18 days.

Task 5: Adversarial Test: What Did the Model Miss? (10 min)

A regression asks only one question: *is this cost unusual for this many days?* Think like someone trying to sneak a bad report past it. Run the notebook's two extra checks (duplicates and days-vs-dates) and answer:

1. Which reports did these checks find that the regression did **not** flag? Why couldn't regression catch them?

   Check A found a duplicate submission: TR-2039 and TR-2076, both K. Brooks, Chicago, Trade show, March 9-12, 4 days, $1,631.40. The cleaned-data line predicts about $1,602.53 for 4 days, so each copy has a residual of only about $29 and sits right on the line. Check B found two reports with inflated days. TR-2018 (T. Nguyen) claimed 7 days, but the dates cover only 3, and its cost of $2,461.85 is close to the $2,406.92 the line predicts for 7 days (a residual of about $55). That report was never flagged. TR-2019 claimed 9 days for a single-day trip (January 27 to January 27). The regression did flag it, but for the wrong reason: it looked too cheap for 9 days, when the real problem is the inflated day count.

   Regression couldn't catch these because it only asks whether a cost is unusual for the days claimed, and it takes days_claimed as true. A duplicate has a normal amount, and each copy looks fine because the model examines rows one at a time and never compares one report with another. Inflated days make a padded cost look normal, since the model is checking the cost against the wrong number of days.

2. For the days mismatch, use your Task 3 equation to calculate what the report should have cost for the actual number of days. How much might Meridian be overpaying?

   Using the Task 3 equation, predicted cost = $530.01 + $268.13 × days. TR-2018: for the actual 3 days, the model predicts about $1,334.40. The company paid $2,461.85, so it may have overpaid by about $1,127.45. TR-2019: for the actual 1 day, the model predicts about $798.14. The company paid $742.10, which is about $56 under the prediction. The cost itself looks normal for a 1-day trip, so the problem here is the false day count, not an overpayment. The two checks together point to roughly $1,127 of possible overpayment from the mismatch, almost all of it from TR-2018. The duplicate adds another $1,631.40 if the second submission was paid.

3. If you were designing Meridian's expense review, where would you use regression and where would you use simple rules? Why do you need both?

   I would use simple rules for things that are always wrong or checkable exactly, such as policy caps, negative totals, duplicate submissions, and days that don't match the dates, because they are cheap, certain, and easy to explain to an employee. I would use regression to catch amounts that break no written rule but are unusual for the trip, such as a normal-looking report that costs three times what a 1-day trip should. You need both because each blind spot is the other's strength: rules catch the cheating and mistakes that look normal, and the model catches the odd amounts that no rule anticipated.

Task 6: Fact-Check an AI-Drafted Audit Memo (15 min)

Part A: Check the draft (12 min)

Meridian's accounts payable team pasted the flagged reports into an AI assistant and asked it to draft a memo to the Controller. The draft is below (it also appears in the notebook). **It reads well, and it contains several errors.** Find them before the Controller acts on it.

> **MEMO (AI-generated draft)**
>
> **To:** Controller, Meridian Fixtures Inc.
>
> **From:** AI Audit Assistant
>
> **Re:** Travel expense reports flagged for review, January–June 2026
>
> **Summary.** We fit a linear regression of total trip cost on days claimed for all travel expense reports submitted from January to June 2026. After removing two reports that violate travel policy (TR-2128 and TR-2120), the model flagged six reports as unusual. The model's R-squared of 0.95 shows that it predicts trip costs with a high degree of accuracy, so these flags can be acted on with confidence.
>
> **TR-2035 (M. Chen, Monterrey, MX, $4,185.60).** This is the largest positive outlier, $2,047 above the expected cost for a 6-day trip. Costs this far above the norm strongly suggest fraudulent padding. **Recommendation: Reject.**
>
> **TR-2060 (M. Chen, Nashville, TN, $2,968.75).** This 2-day customer visit cost $1,902 more than expected, nearly three times the predicted amount. **Recommendation: Request itemized receipts before approving.**
>
> **TR-2081 (J. Alvarez, Denver, CO, $3,712.30).** This 5-day trip is $1,842 above expected and exceeds Meridian's $150-per-day meal allowance. **Recommendation: Request itemized receipts before approving.**
>
> **TR-2084 (R. Osei, Phoenix, AZ, $2,500.00).** Mr. Osei has confirmed that this amount was a flat conference registration fee. **Recommendation: Approve.**
>
> **TR-2019 (T. Nguyen, Columbus, OH, $742.10).** This 9-day trip came in $2,201 under the expected cost, which shows excellent cost discipline. **Recommendation: Approve, and consider recognizing the employee.**
>
> **Conclusion.** All other reports fall within the expected range and can be approved without further review.

Run the notebook's fact-check cell, then answer:

1. On the answer sheet, list **every error** you can find. For each one, write what the memo says, what the data actually shows, and which notebook output or policy item proves it. Look for wrong numbers, invented facts, wrong conclusions, and anything important the memo leaves out.

   On Answer Sheet

2. If the Controller followed the memo exactly as written, which single error would cost Meridian the most, in money or in trust? Why?

   The costliest error is #6, the conclusion that every unflagged report can be approved without further review. If the Controller followed it, Meridian would pay the duplicate TR-2076 ($1,631.40) and the padded TR-2018 (about $1,127.45 over), which is roughly $2,759 in avoidable payments from just the two problems I found. It would also tell reviewers to stop checking the exact kinds of problems the regression can't see. The cost in trust may be larger than the money, because the Controller would be relying on a verdict that was wrong, and it would show that no one checked the model's blind spots. The runner-up is #4, since rejecting TR-2035 would accuse M. Chen of fraud over a trip that is probably legitimate. That would hurt trust, but it affects one employee's reimbursement, while #6 lets a whole category of problems through.

Part B: Generate your own AI memo (8 min)

Copy the flagged-reports table printed by the notebook and paste it into an AI assistant with this prompt:

> *"You are an internal auditor at a building-hardware company. Using the table below of travel expense reports flagged by a regression model (cost vs. trip days), write a short memo to the Controller recommending which reports to approve, question, or reject, and why."*

1. Paste your AI's memo on the answer sheet.

   On Answer Sheet
2. Did your AI make any of the same **kinds** of mistakes you found in Part A? Give one example, or explain what it did better.

   On Answer Sheet
3. Write one rule your team would follow before sending any AI-drafted audit memo to a manager.

   On Answer Sheet

Optional Bonus

Run the bonus cell to refit without the flagged reports. R-squared jumps again. Is it fair to report the higher R-squared to management as "how good the model is"? Why or why not?

No, it is not fair to report this higher R-squared to management. The 0.946 comes from removing the five reports the model fit worst, so it measures the trimmed data, not the model's real performance. Removing the policy violations was justified because they were errors, but removing reports just for being far from the line guarantees a higher R-squared. Management should be given 0.677, with a note that the model only uses days and misses problems like duplicates and inflated day counts.

Answer Sheet

Task 3: Before and After Cleaning

	All data (Task 2)	Cleaned data (Task 3)
Slope	$256.32 per extra day	$268.13 per extra day
Intercept	654.15	530.01
R-squared	0.199	0.677
Rows used	129	127

Task 4: Investigation Table

Action: **Approve**, **Ask the employee**, or **Reject**.

Report ID	Days	Cost	Residual	Likely explanation (cite evidence)	Action
TR-2019	9 Days	$742.10	$2,943	Cost is too little for 9 days, The days claimed were either overstated or the costs were billed elsewhere.	Ask employee
TR-2035	6 Days	$4,185.60	$2,139	International travel is more expensive than domestic. Most likely legitimate.	Approve (after checking airfare receipt)
TR-2060	2 Days	$2,968.75	$1,066	Nearly 3 times the normal cost for short domestic flight. Could be last minute flight or something was upgraded. Could also be longer than a 2-day stay.	Ask employee
TR-2081	5 Days	$3,712.30	$1,871	Training could include a course or a registration fee.	Ask employee (confirm that the fee is itemized)
TR-2084	1 Day	$2,500	$798	About 3 times the normal amount for a 1-day trip, amount is also a round number, so it is either an estimate or a lump sum instead of itemized receipts.	Ask the employee (reject if no receipts)

Task 5: What the Model Missed

Report ID(s)	Problem	Why regression missed it
TR-2039 and TR-2076	Duplicate Submission	The cleaned-data line predicts about $1,602.53 for 4 days, so each copy has a residual of only about $29 and sits right on the line.
TR-2018 and TR-2019	Reports with Inflated Days	TR-2018 (T. Nguyen) claimed 7 days, but the dates cover only 3, and its cost of $2,461.85 is close to the $2,406.92 the line predicts for 7 days (a residual of about $55). That report was never flagged. TR-2019 claimed 9 days for a single-day trip (January 27 to January 27). The regression did flag it, but for the wrong reason: it looked too cheap for 9 days, when the real problem is the inflated day count.
		Regression couldn't catch these because it only asks whether a cost is unusual for the days claimed, and it takes days_claimed as true. A duplicate has a normal amount, and each copy looks fine because the model examines rows one at a time and never compares one report with another. Inflated days make a padded cost look normal, since the model is checking the cost against the wrong number of days.

Estimated overpayment on the days-mismatch report: $1,127.45

Task 6, Part A: Errors in the AI-Drafted Memo

#	What the memo says	What the data actually shows	Proof (notebook output or policy item)
1	The model flagged six reports.	It flagged five: TR-2019, TR-2035, TR-2060, TR-2081, and TR-2084. The memo itself only discusses five.	Task 4: “Reports flagged: 5”
2	R-squared is 0.95, so the flags can be acted on with confidence.	R-squared is 0.677.	Task 3 output and Task 6 check cell (0.677)
3	TR-2019 is a 9-day trip showing "excellent cost discipline." Approve and recognize the employee.	The trip dates are 2026-01-27 to 2026-01-27, so it was 1 day. The low residual comes from the inflated day count, not from thrift. The recommendation should be to ask the employee.	Task 5 Check B and the Task 6 date table (days_from_dates = 1)

#	What the memo says	What the data actually shows	Proof (notebook output or policy item)
4	TR-2035 strongly suggests fraudulent padding. Reject.	The dates match the 6 days claimed. The destination is Monterrey, MX, an international trip, so a higher cost is plausible, and the model doesn't know about destination. The recommendation should be to approve after a receipt check or ask the employee.	Task 6 date table (6 = 6) and Task 4 table (destination)
5	Mr. Osei has confirmed that the $2,500 was a flat conference registration fee. Approve.	Nothing in the data shows any confirmation, and the purpose is Customer visit, not a conference. It was a 1-day trip (2026-06-03 to 2026-06-03) with a predicted cost of $798.14, a residual of $1,701.86, and a suspiciously round amount. The recommendation should be to ask the employee for receipts.	Task 4 table; Task 6 date table
6	All other reports fall within the expected range and can be approved without further review.	The regression can't see duplicates or inflated days, so unflagged reports can still be problems.	Task 5 outputs
7	Left out the duplicate submission.	TR-2039 and TR-2076 are the same K. Brooks Chicago trade show, March 9–12, $1,631.40 each. Both sat on the line, so neither was flagged.	Task 5 Check A
8	Left out TR-2018.	It claims 7 days, but the dates cover only 3. Its cost of $2,461.85 is close to the 7-day prediction, so the model never flagged it. At Task 3, 3 days should cost about $1,334.40, so the possible overpayment is about $1,127.45.	Task 5 Check B
9	Left out what happens to the two removed reports.	TR-2120 (-$1,186.40) and TR-2128 ($13,482.50) should go back to the employee or approver to be corrected, since the true amounts may be ordinary. R. Osei filed both TR-2128 and TR-2084, which is a reason to take a closer look, not proof of a problem.	Task 3 errors table & Task 4 table

Costliest error and why

The costliest error is #6, the conclusion that every unflagged report can be approved without further review. If the Controller followed it, Meridian would pay the duplicate TR-2076 ($1,631.40) and the padded TR-2018 (about $1,127.45 over), which is roughly $2,759 in avoidable payments from just the two problems I found. It would also tell reviewers to stop checking the exact kinds of problems the regression can't see. The cost in trust may be larger than the money, because the Controller would be relying on a verdict that was wrong, and it would show that no one checked the model's blind spots. The runner-up is #4, since rejecting TR-2035 would accuse M. Chen of fraud over a trip that is probably legitimate. That would hurt trust, but it affects one employee's reimbursement, while #6 lets a whole category of problems through.

Task 6, Part B: Your Own AI Memo

Paste your AI's memo here

MEMORANDUM
To: Controller
From: Internal Audit
Date: October 4, 2026
Re: Travel expense reports flagged by the cost-vs-days regression model
Summary
The model flagged five reports. I recommend approving one, questioning three, and rejecting one as submitted. The model predicts cost from trip length only, so a flag is a prompt for review, not evidence of wrongdoing. It cannot account for airfare, international travel, or conference fees.

Report	Employee	Claimed	Predicted	Recommendation
TR-2019	T. Nguyen	$742.10	$2,943.17	**Approve** (correct the days field)
TR-2035	M. Chen	$4,185.60	$2,138.78	**Question**
TR-2060	M. Chen	$2,968.75	$1,066.27	**Question** (priority)
TR-2081	J. Alvarez	$3,712.30	$1,870.65	**Question**
TR-2084	R. Osei	$2,500.00	$798.14	**Reject as submitted**

Approve: TR-2019 (Nguyen, Columbus)
This flag is a data error, not an overspend. The report shows a single-day trip (Jan 27) but claims 9 days. The model expected about $2,943 for nine days, so the large negative residual is an artifact. At $742 for one day, the cost is reasonable for a supplier audit. I recommend approving it and having the days field corrected so it doesn't distort future model runs.
Question

- **TR-2060 (Chen, Nashville):** This claims $2,969 for two days, about $1,484 per day and nearly 3x the prediction, with no obvious reason for a domestic customer visit. It is also Chen's second flag, so I recommend requesting full itemization and receipts.
- **TR-2035 (Chen, Monterrey):** The trip is 6 days, matching the dates, and costs about 2x the prediction. International airfare could explain much of the gap, so this is likely approvable once receipts confirm it. I suggest reviewing it together with TR-2060, since the two reports total roughly $3,950 above predicted.
- **TR-2081 (Alvarez, Denver sales training):** The dates match the 5 days claimed, and the cost is about 2x the prediction. Registration or tuition fees may be included, and those should be coded separately from travel. I recommend requesting an itemized breakdown before approving.

Paste your AI's memo here
Reject as submitted: TR-2084 (Osei, Phoenix) This claims exactly $2,500.00 for a one-day customer visit, over 3x the prediction and the highest cost per day in the set. A perfectly round total suggests an estimate or lump sum rather than actual receipts. I recommend returning it for resubmission with itemized receipts. This is a documentation rejection, not a finding of misconduct, and we should approve whatever the receipts support. **Exposure and next steps** The four over-predicted reports total $13,366.65 claimed against $5,873.84 predicted, a gap of about $7,493. Not all of it will be unsupported, so the amount actually at risk depends on what the receipts show. I suggest a 10-business-day deadline for documentation. I also suggest a brief look at Chen's other recent claims, given two flags in one quarter, and a refresh of the model with destination and trip type (domestic vs. international) to reduce false positives.

Same kind of mistake as Part A? (example)	Your team's rule for AI-drafted memos
My AI did better than the draft memo on the biggest problems. It reports the correct count of five flagged reports and never quotes an inflated R-squared. It says a flag is "a prompt for review, not evidence of wrongdoing", so it doesn't call anyone a fraud. It also caught that TR-2019 claims 9 days for a trip whose dates cover one day, and it didn't invent evidence like the "Mr. Osei has confirmed". Its dollar figures check out: the $13,366.65 claimed, the $5,873.84 predicted, and the roughly $7,493 gap all match my Task 4 output. For TR-2084 it says "documentation rejection, not a finding of misconduct" and asks for receipts. It did make one mistake of having an unsupported claim. It says Chen has "two flags in one quarter", but TR-2035 is dated February 2-7 and TR-2060 is April 21-22, which are different calendar quarters. The pattern is real, but the wording goes beyond what the dates show. It also leaves out some things, as the draft memo did. It never mentions the duplicate (TR-2039 and TR-2076) or the TR-2018 days mismatch from Task 5, and it doesn't say that unflagged reports can still have problems the regression can't see. It points to Chen's two flags but not to R. Osei, who also filed TR-2128, the report removed in Task 3.	Before any AI-drafted audit memo goes to a manager, a person must trace every number, date, and factual claim back to the notebook output or source data, and delete anything that can't be traced. The reviewer must also check that the memo says what the model cannot see, such as duplicates and days that don't match the dates.